



9400 - Mapping dust and stars at the cosmic star formation peak

Cycle: 4, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Irene Shivaiei (PI) (ESA Member)	Centro de Astrobiologia - CAB
Prof. Caitlin M. Casey (CoI)	University of California - Santa Barbara
Dr. Alexandra Pope (CoI) (US Admin CoI)	University of Massachusetts - Amherst
Dr. Gergo Popping (CoI) (ESA Member)	European Southern Observatory - Germany
Leonor Arriscado (CoI) (ESA Member)	Centro de Astrobiologia - CAB
Dr. Ivanna Langan (CoI) (ESA Member)	Centro de Astrobiologia - CAB
Dr. Naveen A. Reddy (CoI)	University of California - Riverside
Roxana Popescu (CoI)	University of Massachusetts - Amherst
Dr. Leindert Boogaard (CoI) (ESA Member)	Universiteit Leiden
Dr. Melanie Kaasinen (CoI)	Australian National University
Dr. Marko Shuntov (CoI) (ESA Member)	Cosmic Dawn Center, Niels Bohr Institute
Dr. Santosh Harish (CoI)	Space Telescope Science Institute
Dr. Maximilien Franco (CoI) (ESA Member)	Universite Paris-Saclay

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		MIRI Imaging	(1) 24763
	2		MIRI Imaging	(2) 3324-5094
	3		MIRI Imaging	(3) 13701
	4		MIRI Imaging	(4) 5814
	5		MIRI Imaging	(5) 9393

ABSTRACT

We propose joint high-resolution ALMA Band-7 (ID 2025.1.01403.S) and JWST/MIRI observations of six main-sequence galaxies at $z \sim 2.3$ to map cold dust, warm dust, and stellar mass at matched resolution for a kpc-scale study of dust-star geometry and galaxy growth during the peak of cosmic star formation. These galaxies are representative of the typical $z \sim 2$ galaxies, and have lower masses and more extensive ancillary data than any other similar study to date. The targets have integrated molecular and ionized gas observations, revealing key properties of their gaseous ISM. They have multi-band NIRC2 imaging, providing maps of unobscured emission from young stars. With the new observations, we aim to map cold dust (ALMA B7), warm dust/PAHs (MIRI 10-21 μ m), and stellar mass (MIRI 7.7 μ m) at matching kpc-scale resolutions to (1) study in-situ galaxy growth by disentangling dust attenuation from intrinsic stellar structure, (2) test for dust temperature gradients by comparing dust size and concentration across wavelengths, and (3) constrain galaxy-wide PAH and dust formation scenarios. With this dataset, we will uncover how dust geometry shapes our view of galaxy evolution in typical galaxies at cosmic noon.

OBSERVING DESCRIPTION

The planned MIRI imaging in the F2100W, F1500W, F1000W, and F770W filters is designed to study resolved stellar mass (F770W traces rest-frame ~ 2 -3 μ m) and PAH and warm dust emission (F1000W to F2100W tracing rest-frame ~ 3 -7 μ m). Existing surveys (PIDs 1727, 1837, 5893) already provide some of this data. We request only the missing bands. In short, two targets (IDs 3284,9393) have full F770W from PID 1727 (see duplications below). One targets (ID 13701) have F1000W and F2100W from PID 5893.

Exposure times are calculated to ensure the detection of MIR emission in galaxies with lowest star formation rates (SFR ~ 30 Msun/yr), based on flux estimates from the MIRI SMILES survey for galaxies at $z \sim 2$ -2.5. According to ETC and consistent with the existing observations from MIRI multi-band surveys of SMILES, PRIMER, CEERS, the requested exposure times will achieve a SNR of 10 at flux levels of 0.7, 1, and 3 microJy at 10, 15, and 21 micron, respectively, ensuring robust detection of extended MIR emission in the lowest-SFR galaxies of the sample. MIRI 7.7 μ m exists for 2 (out of 6) sources from the COSMOS-Web program. For the rest missing 7.7 μ m data, we request F770W imaging at a matching depth for data consistency (SNR=10 at 0.3uJy).

Pointings and PA restrictions: 5 MIRI pointings are strategically allocated to cover the 6 galaxies in the sample, as MIRI's small field of view can only cover one or two galaxies per pointing. The survey tiles and position angles (PAs) are optimized to maximize the coverage of the sample while ensuring efficient use of telescope time.

JWST Proposal 9400 (Created: Tuesday, February 3, 2026, 9:00:22AM Eastern Standard Time) - Overview

Justification of Duplications: PID 5893 has planned observations in the location of 3324+5094 (one MIRI pointing). Given that PID 5893 is a survey and MIRI is used in parallel, coverage of these sources depend on the final PA of the pointings. We therefore include them in the APT in case they end up not being covered by final mosaics of PID 5893. Additionally, even though target 5094 is in the PID 1727 F770W coverage, it is on the edge of the footprint (rms =0.11) and not fully covered (rms of the rest of the field is 0.02). Therefore, we request F770W observations on this source as well to reach the desired depth.

Proposal 9400 - Targets - Mapping dust and stars at the cosmic star formation peak

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	24763	RA: 10 00 13.6084 (150.0567017d) Dec: +02 26 4.79 (2.43466d) Equinox: J2000 <i>Comments:</i> <i>Category=Galaxy</i> <i>Description=[Field galaxies]</i>		
	(2)	3324-5094	RA: 10 00 34.9003 (150.1454179d) Dec: +02 13 26.27 (2.22396d) Equinox: J2000 <i>Comments:</i> <i>Category=Galaxy</i> <i>Description=[Field galaxies]</i>		
	(3)	13701	RA: 10 00 27.2359 (150.1134829d) Dec: +02 19 10.71 (2.31964d) Equinox: J2000 <i>Comments:</i> <i>Category=Galaxy</i> <i>Description=[Field galaxies]</i>		
	(4)	5814	RA: 10 00 39.8827 (150.1661779d) Dec: +02 14 37.85 (2.24385d) Equinox: J2000 <i>Comments:</i> <i>Category=Galaxy</i> <i>Description=[Field galaxies]</i>		
	(5)	9393	RA: 10 00 36.2288 (150.1509533d) Dec: +02 16 48.44 (2.28012d) Equinox: J2000 <i>Comments:</i> <i>Category=Galaxy</i> <i>Description=[Field galaxies]</i>		

Proposal 9400 - Observation 1 - Mapping dust and stars at the cosmic star formation peak

Observation	Proposal 9400, Observation 1										Tue Feb 03 14:00:22 GMT 2026
	Diagnostic Status: Warning										
Diagnostics	Observing Template: MIRI Imaging										
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	24763	RA: 10 00 13.6084 (150.0567017d) Dec: +02 26 4.79 (2.43466d) Equinox: J2000								
Template	Comments: Category=Galaxy Description=[Field galaxies]										
	Subarray										
Dithers	FULL										
	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
Spectral Elements	1	CYCLING	1	4		1	1			LARGE	
	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	60	3	1	Dither 1	4	12	2020.229	
	2	F1000W	FASTR1	85	1	1	Dither 1	4	4	943.514	
	3	F1500W	FASTR1	80	2	1	Dither 1	4	8	1787.126	
4	F2100W	FASTR1	48	4	1	Dither 1	4	16	2164.531		

Proposal 9400 - Observation 2 - Mapping dust and stars at the cosmic star formation peak

Observation	Proposal 9400, Observation 2										Tue Feb 03 14:00:22 GMT 2026
	Diagnostic Status: Warning										
Observing Template: MIRI Imaging											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	3324-5094	RA: 10 00 34.9003 (150.1454179d) Dec: +02 13 26.27 (2.22396d) Equinox: J2000								
Comments: Category=Galaxy Description=[Field galaxies]											
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4		1	1			LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	60	3	1	Dither 1	4	12	2020.229	
	2	F1000W	FASTR1	85	1	1	Dither 1	4	4	943.514	
	3	F1500W	FASTR1	80	2	1	Dither 1	4	8	1787.126	
	4	F2100W	FASTR1	48	4	1	Dither 1	4	16	2164.531	

Proposal 9400 - Observation 3 - Mapping dust and stars at the cosmic star formation peak

Observation	Proposal 9400, Observation 3										Tue Feb 03 14:00:22 GMT 2026
	Diagnostic Status: Warning										
Diagnostics	Observing Template: MIRI Imaging										
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	13701	RA: 10 00 27.2359 (150.1134829d) Dec: +02 19 10.71 (2.31964d) Equinox: J2000								
Template	Comments: Category=Galaxy Description=[Field galaxies]										
	Subarray										
Dithers	FULL										
	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
Spectral Elements	1	CYCLING	1	4		1	1			LARGE	
	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	60	3	1	Dither 1	4	12	2020.229	
	2	F1500W	FASTR1	80	2	1	Dither 1	4	8	1787.126	

Proposal 9400 - Observation 4 - Mapping dust and stars at the cosmic star formation peak

Observation	Proposal 9400, Observation 4										Tue Feb 03 14:00:22 GMT 2026
	Diagnostic Status: Warning										
Observing Template: MIRI Imaging											
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(4)	5814	RA: 10 00 39.8827 (150.1661779d)								
			Dec: +02 14 37.85 (2.24385d)								
			Equinox: J2000								
Template	Comments: Category=Galaxy Description=[Field galaxies]										
	Subarray										
FULL											
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4		1	1			LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	60	3	1	Dither 1	4	12	2020.229	
	2	F1000W	FASTR1	85	1	1	Dither 1	4	4	943.514	
	3	F1500W	FASTR1	80	2	1	Dither 1	4	8	1787.126	
	4	F2100W	FASTR1	48	4	1	Dither 1	4	16	2164.531	

Proposal 9400 - Observation 5 - Mapping dust and stars at the cosmic star formation peak

Observation	Proposal 9400, Observation 5										Tue Feb 03 14:00:22 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(5)	9393	RA: 10 00 36.2288 (150.1509533d) Dec: +02 16 48.44 (2.28012d) Equinox: J2000								
	Comments: Category=Galaxy Description=[Field galaxies]										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4		1	1			LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F1000W	FASTR1	85	1	1	Dither 1	4	4	943.514	
	2	F1500W	FASTR1	80	2	1	Dither 1	4	8	1787.126	
	3	F2100W	FASTR1	48	4	1	Dither 1	4	16	2164.531	
Special Requirements	Aperture PA Range 81.83544897 to 145.83544897 Degrees (V3 77.0 to 141.0)										